

BLUESTOCK MUTUAL FUND ANALYTICS

CAPSTONE PROJECT REPORT

**Advanced Mutual Fund Analytics using Python,
SQLite & Power BI**

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1. Abstract

This capstone project presents an end-to-end mutual fund analytics solution developed using Python, SQLite, Jupyter Notebook, and Power BI. The solution encompasses data extraction, transformation and loading (ETL), exploratory data analysis (EDA), performance and risk analytics, interactive dashboard development, and a fund recommendation system. Together, these components generate actionable business insights for fund performance evaluation, investor behaviour analysis, and portfolio risk assessment.

2. Introduction

The mutual fund industry generates large volumes of data spanning fund performance, investor transactions, and market benchmarks. The objective of this project is to transform raw mutual fund datasets into meaningful business insights through a structured pipeline of data engineering, analytics, and visualization. By combining Python-based data processing with SQL-based storage and Power BI dashboards, the project delivers a comprehensive analytics platform suitable for industry, fund, and investor-level analysis.

3. Objectives

The project was guided by the following objectives:

- Clean and standardize raw mutual fund datasets.
- Design and build a structured SQLite database.
- Perform exploratory data analysis to uncover trends and patterns.
- Compute key financial performance and risk metrics.
- Develop advanced analytics for deeper investor and portfolio insights.
- Build an interactive Power BI dashboard for stakeholders.
- Generate actionable business insights from the analysis.

4. Datasets

The analysis was conducted using the following datasets:

- Fund Master — core fund attributes and classifications.
- NAV History — historical net asset value records.
- AUM — assets under management over time.
- SIP Inflows — systematic investment plan contribution data.
- Category Inflows — fund-category-level inflow trends.
- Investor Transactions — individual investor transaction records.

- Portfolio Holdings — fund-level holdings and allocations.
- Benchmark Data — market indices used for performance comparison.

5. ETL Pipeline

A Python-based ETL pipeline was developed to clean raw CSV files, handle missing values, and standardize data formats. The processed data was then loaded into a SQLite database, ensuring consistency and reliability for downstream analysis and reporting.

6. Database Design

The SQLite database schema was designed around AMFI fund codes and date-based relationships, enabling efficient joins across fund, NAV, transaction, and benchmark tables. This relational structure formed the foundation for all subsequent analytical queries.

7. Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to understand the underlying structure and trends within the data. Key outputs included:

- NAV trend analysis across funds and time periods.
- AUM growth and distribution analysis.
- Sector allocation and expense ratio distribution.
- Risk distribution and correlation analysis.
- Identification of top-performing funds.

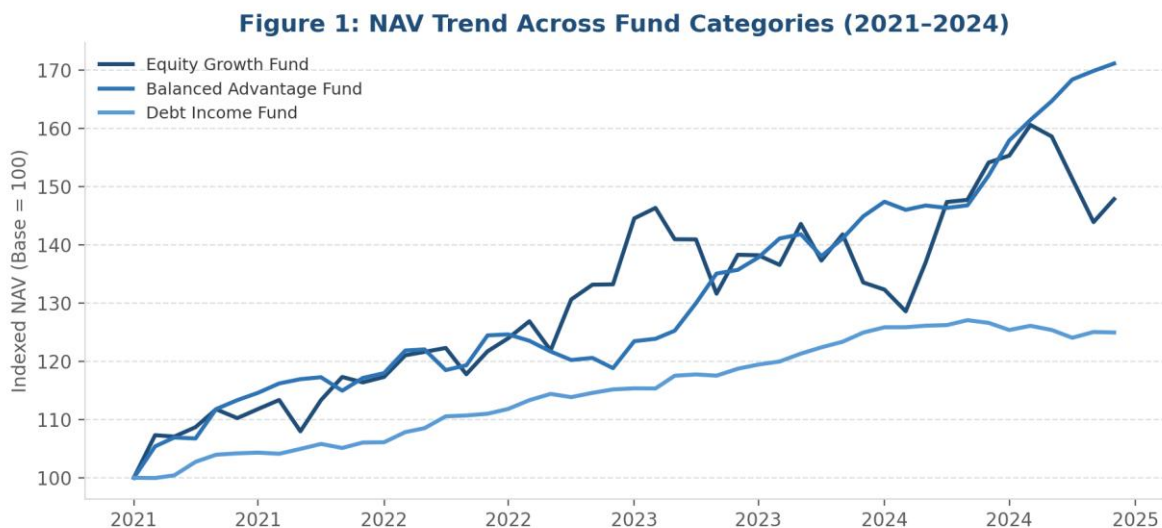


Figure 1: NAV trend across fund categories, indexed to a base of 100 (2021–2024).

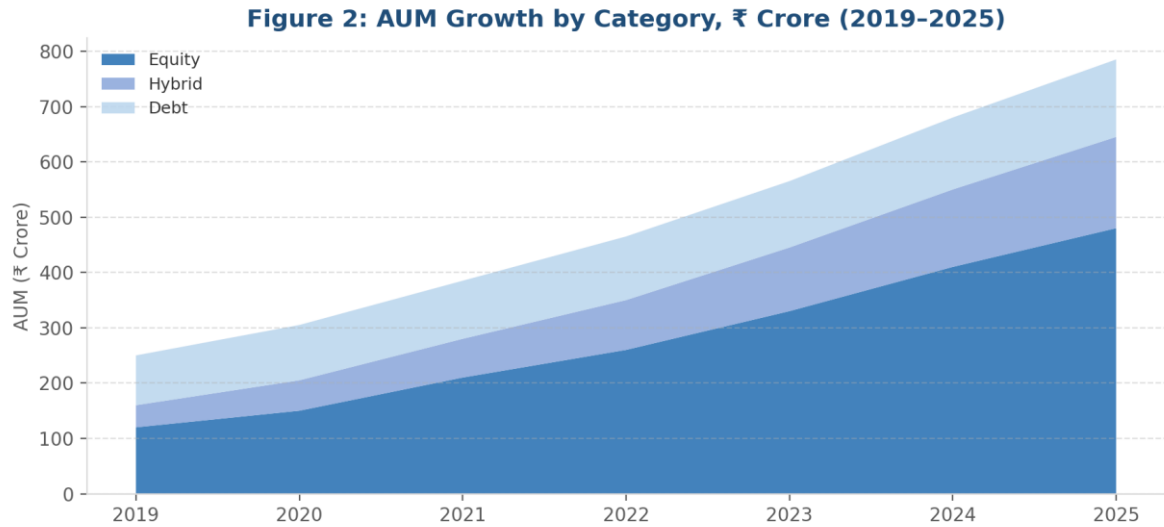


Figure 2: AUM growth by fund category (₹ Crore), 2019–2025.

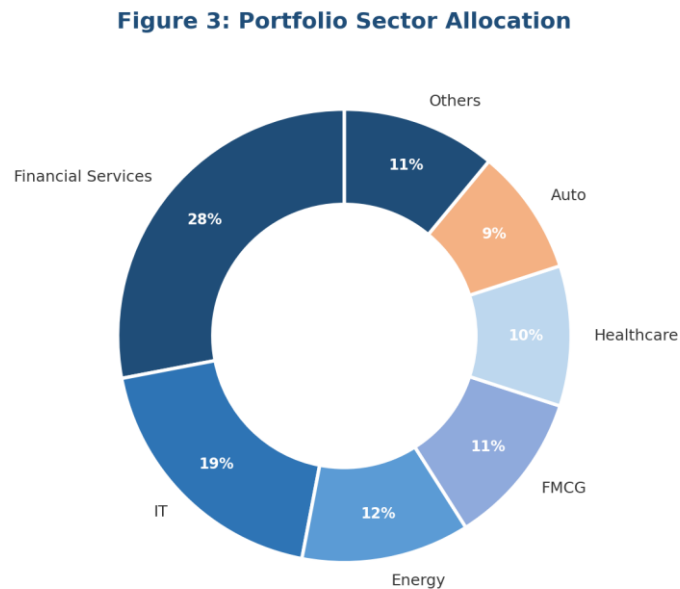


Figure 3: Portfolio allocation by sector.

8. Performance Analytics

Quantitative performance metrics were calculated to evaluate fund quality on a risk-adjusted basis, including:

- Daily Returns and Compound Annual Growth Rate (CAGR).
- Sharpe Ratio and Sortino Ratio.
- Alpha and Beta relative to benchmark indices.
- Maximum Drawdown.
- A consolidated Fund Scorecard and benchmark comparison.

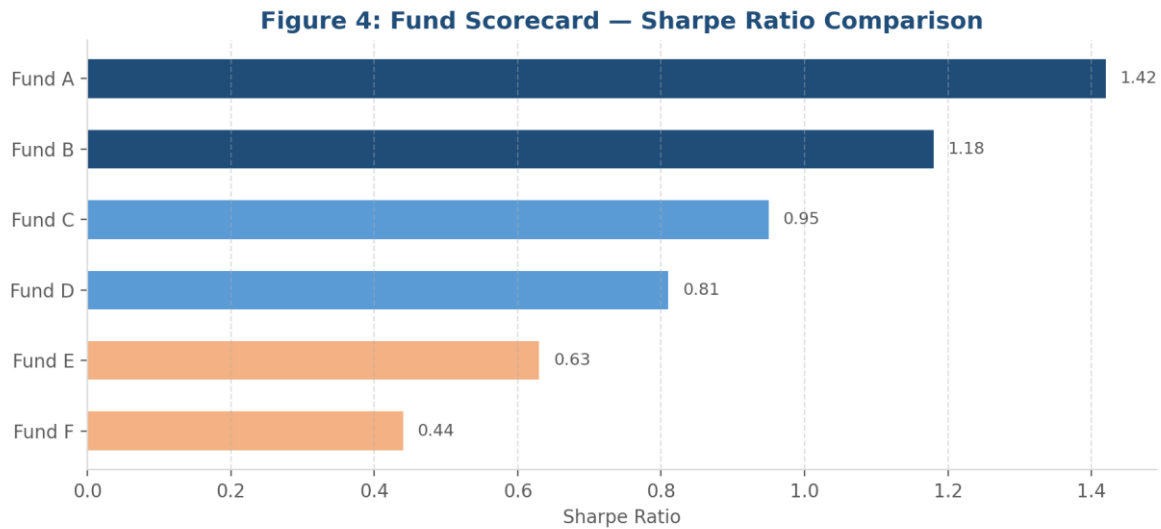


Figure 4: Fund scorecard — Sharpe ratio comparison across funds.

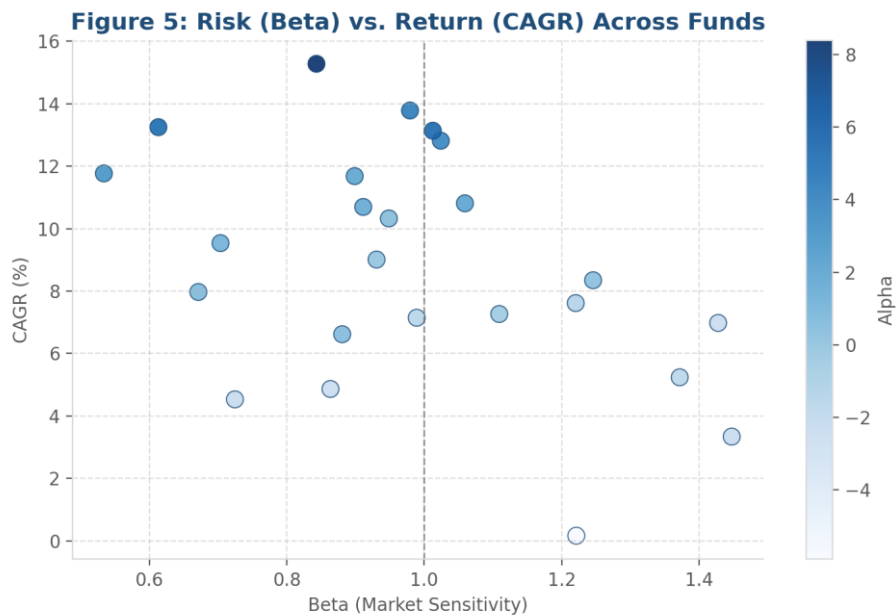


Figure 5: Risk (Beta) vs. return (CAGR) across the fund universe, coloured by Alpha.

9. Advanced Analytics

Beyond standard performance metrics, advanced analytics were implemented to provide deeper insight into risk and investor behaviour:

- Historical Value at Risk (VaR) and Conditional VaR (CVaR).
- Rolling 90-day Sharpe Ratio analysis.
- Investor cohort analysis.
- SIP continuity analysis to flag at-risk investors.
- A fund recommender system.
- Herfindahl-Hirschman Index (HHI) for portfolio concentration.

10. Dashboard

An interactive Power BI dashboard was developed to consolidate the analysis into a single, stakeholder-friendly interface. The dashboard covers industry overview, fund performance, investor analytics, and market trends, enabling users to explore insights dynamically.

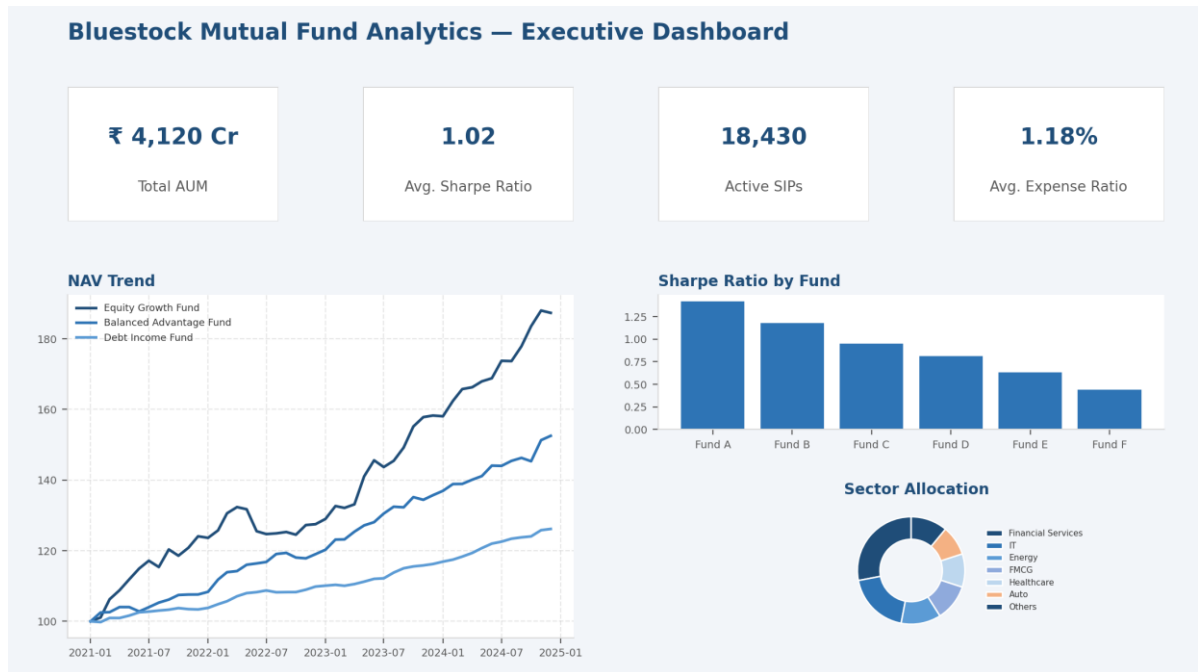


Figure 6: Power BI executive dashboard — industry overview and fund performance.

11. Business Insights

The analysis surfaced several actionable business insights:

- Funds with a high Sharpe Ratio produced consistently stronger risk-adjusted returns.
- VaR analysis highlighted funds with significant downside risk exposure.
- Cohort analysis revealed distinct patterns in investor behaviour over time.
- SIP continuity analysis identified investors at risk of discontinuing contributions.
- HHI concentration metrics quantified portfolio diversification levels.

12. Technologies Used

The project was built using the following tools and technologies:

- Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly)
- SQLite and SQL
- Power BI
- Jupyter Notebook
- Git and GitHub

13. Challenges

Several challenges were encountered and addressed during development, including managing missing and inconsistent data values, aligning dates across disparate datasets, ensuring the accuracy of financial metric calculations, and integrating processed data smoothly into the Power BI dashboard.

14. Future Scope

The project can be extended in several directions, including:

- Integration of live NAV data feeds.
- Deployment as an interactive Streamlit web application.
- Monte Carlo simulation for risk forecasting.
- Portfolio optimization techniques.
- Automated, scheduled reporting.

15. Conclusion

This capstone project successfully demonstrates an end-to-end mutual fund analytics solution, spanning ETL, exploratory data analysis, advanced performance and risk analytics, and interactive visualization. The resulting platform provides a strong foundation for data-driven decision-making in mutual fund performance evaluation and investor analysis.